

From Correlation to Causation: A Causal Analysis of Heart Disease Risk Using Observational Data

A Causal Inference Study Using DAGs, Counterfactual Reasoning, and Treatment Effect Estimation

PROBLEM STATEMENT : Traditional machine learning models in healthcare focus on prediction by learning correlations in data. However, correlation does not imply causation, and relying solely on predictive models can lead to misleading conclusions in high-stakes domains like healthcare.

This project aims to move beyond correlation by estimating causal effects using a structured causal inference framework, enabling intervention-based reasoning rather than passive prediction.

DATASET DESCRIPTION :

This study utilizes a heart disease dataset containing clinical and physiological attributes of patients to analyze factors influencing heart disease risk.

This dataset is widely used as a benchmark in healthcare analytics and machine learning research.

- **Source:**

The dataset is derived from a publicly available heart disease dataset commonly used for predictive and analytical studies in healthcare.

- **Number of Observations:**

The dataset contains approximately ~1000 patient records.

- **Target Variable:**

- target: Binary variable indicating presence (1) or absence (0) of heart disease.

- **Key Features Used in This Study:**

- oldpeak: ST depression induced by exercise relative to rest. It reflects abnormalities in heart function and is treated as the primary exposure (treatment) variable in this analysis.

- age: Age of the patient (in years), considered as a potential confounder influencing both oldpeak and heart disease risk.

- **Role of Variables in Causal Framework:**

- Exposure (Treatment): oldpeak

- Outcome: target (heart disease)

- Confounder: age

- **Data Characteristics:**

- Observational (non-experimental) dataset

- Binary classification problem

- Presence of subgroup imbalance (notably in older age groups)

- Potential confounding relationships between variables

- **Motivation for Feature Selection:**

The analysis focuses on understanding whether oldpeak has a causal effect on heart disease and how this effect varies across age groups, making age a critical adjustment variable.

Why this matters?

Unlike traditional machine learning projects that focus solely on prediction accuracy, this project emphasizes causal reasoning. It demonstrates how models can be used not just to predict outcomes, but to answer deeper questions such as:

- What happens if we intervene?
- For whom does a treatment work best?
- Where is intervention most effective?

This shift from prediction to causation is critical for real-world decision-making, particularly in healthcare and policy systems.

This project bridges the gap between predictive modeling and actionable decision-making, which is essential in high-stakes domains like healthcare.

IMPLEMENTATION SUMMARY:

This study follows a causal inference framework to move beyond correlation-based analysis and estimate the causal effect of clinical variables on heart disease outcomes.

1. Causal Graph Construction:

A Directed Acyclic Graph (DAG) was constructed using domain knowledge to represent assumed relationships between variables. Age was identified as a confounder influencing both oldpeak and heart disease.

2. Confounder Adjustment:

To obtain unbiased causal estimates, logistic regression models were trained with and without adjustment for age. This allowed comparison between naive (unadjusted) and adjusted estimates, highlighting the impact of confounding.

3. Model Specification:

A logistic regression model was used due to the binary nature of the target variable. The model estimates log-odds, which are transformed into probabilities using the sigmoid function.

4. Intervention Simulation (do-operator):

To estimate causal effects, interventions were simulated by fixing values of oldpeak across the dataset while keeping other variables unchanged. This approximates the $\text{do}(\text{oldpeak} = x)$ operation.

5. Treatment Effect Estimation:

- Average Treatment Effect (ATE) was computed to measure the overall impact of changes in oldpeak.
- Individual Treatment Effect (ITE) was estimated for specific patients to capture personalized responses.

6. Heterogeneity Analysis:

The effect of oldpeak was analyzed across different age groups to understand variation in treatment effects (HTE). Heatmaps and subgroup analysis were used to identify regions of strong and weak effects.

7. Interaction Analysis:

An interaction term (oldpeak $\times$ age) was included in the model to test whether age modifies the effect of oldpeak on heart disease.

8. Model Robustness Check:

A Random Forest model was used as a non-linear alternative to compare causal response patterns and validate the consistency of findings.

9. Visualization Techniques:

Multiple visualizations including scatter plots, probability curves, 3D surfaces, contour plots, and heatmaps were used to interpret relationships and validate causal behavior.

Figure 1: Comparison between naive and confounder-adjusted models, highlighting the impact of age as a confounder.

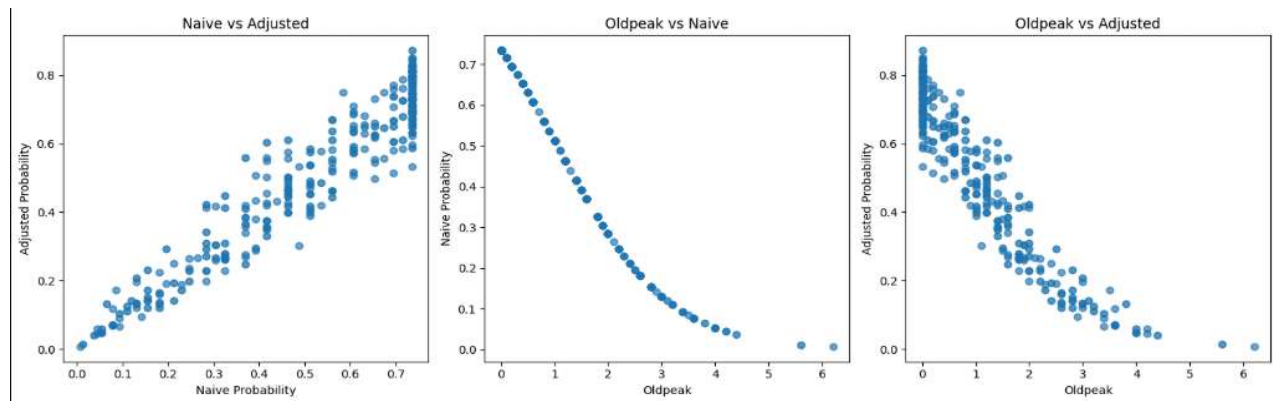
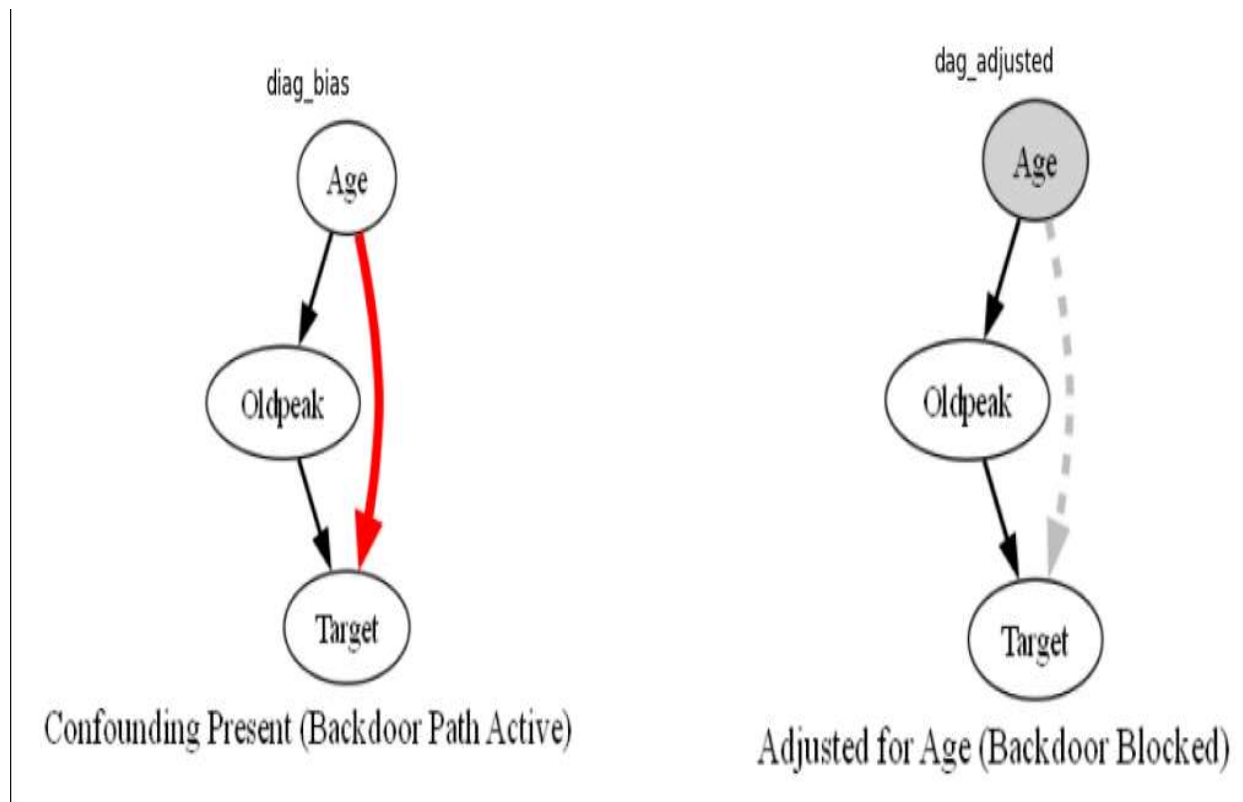


Figure 2: Causal graph illustrating confounding bias and its removal through adjustment.



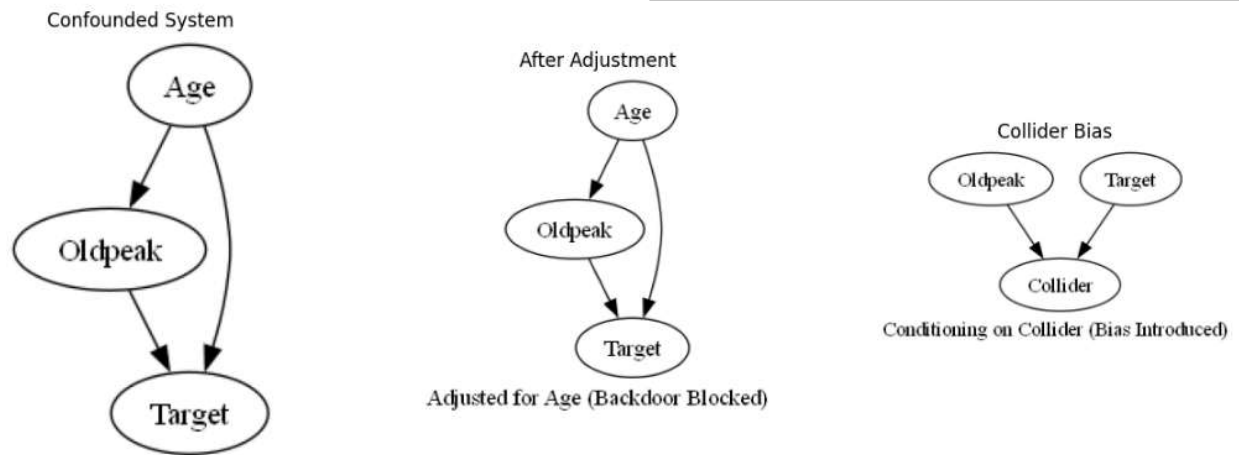
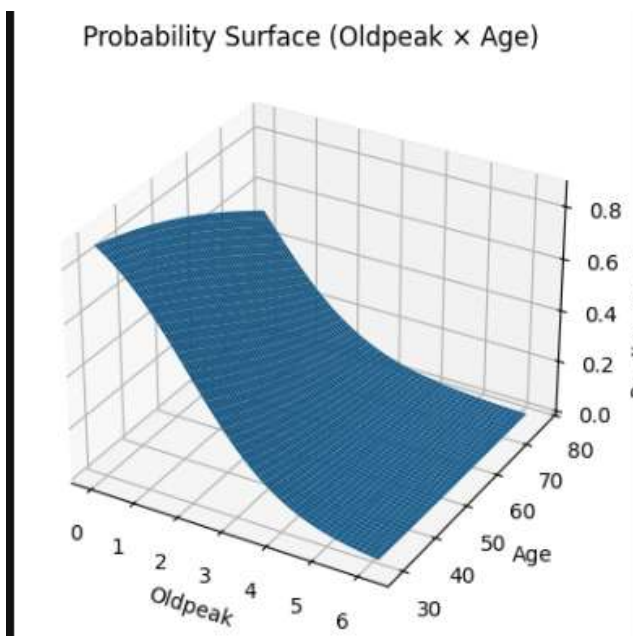
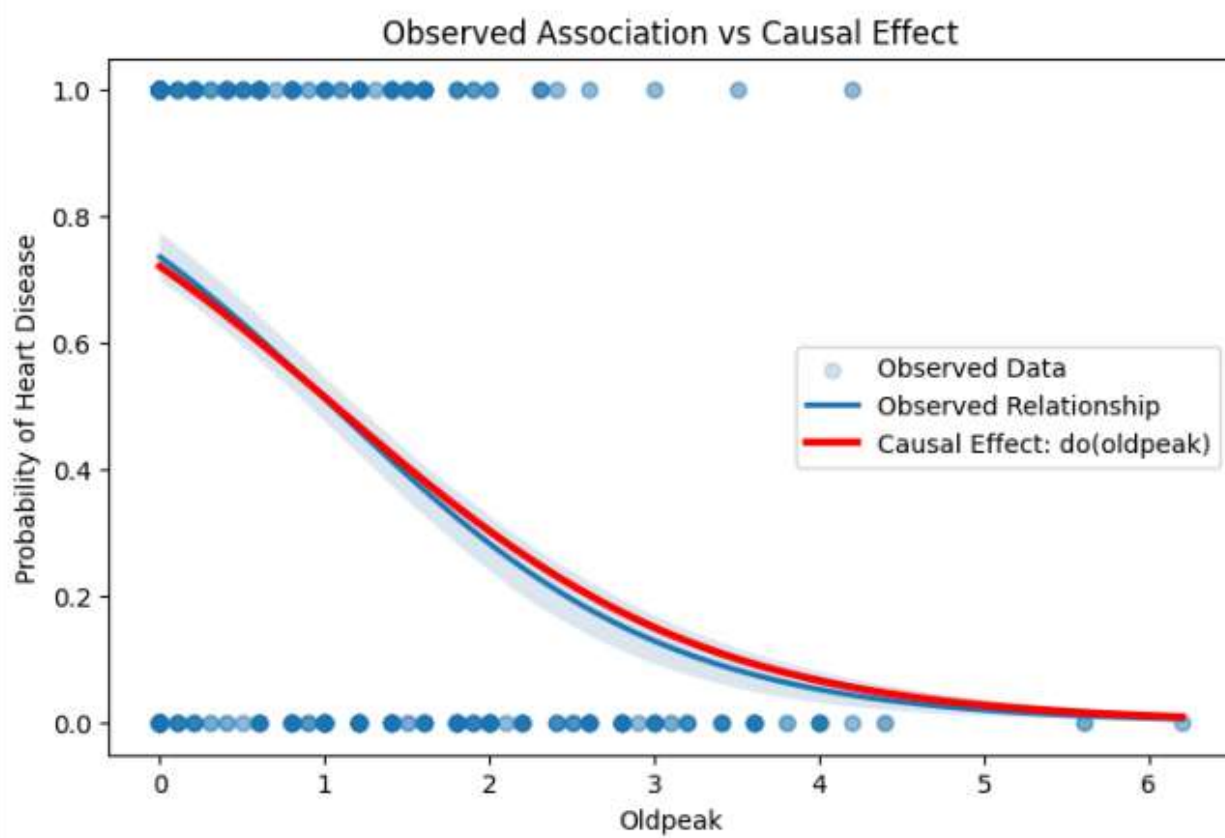
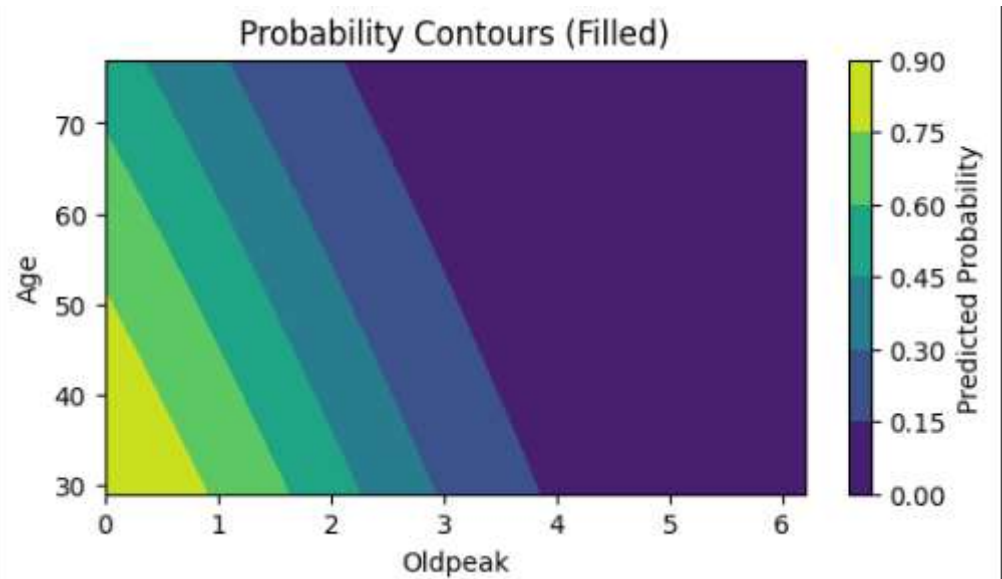
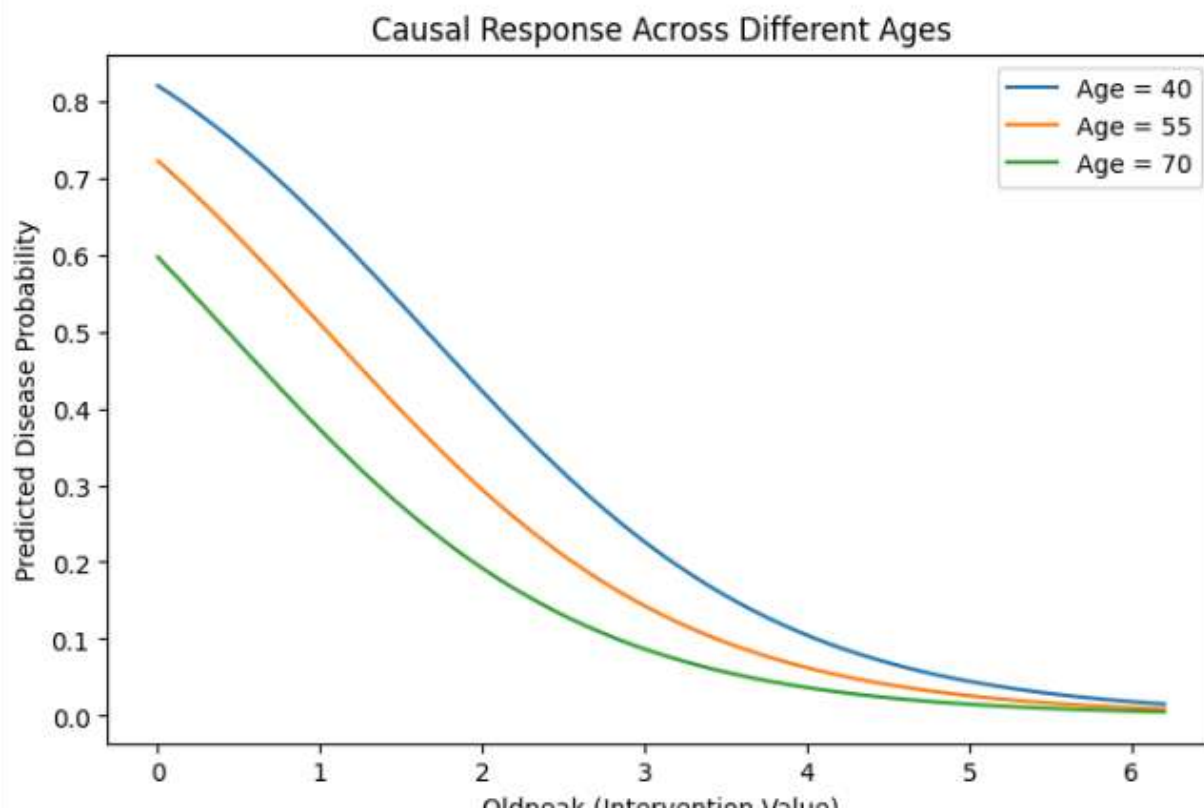
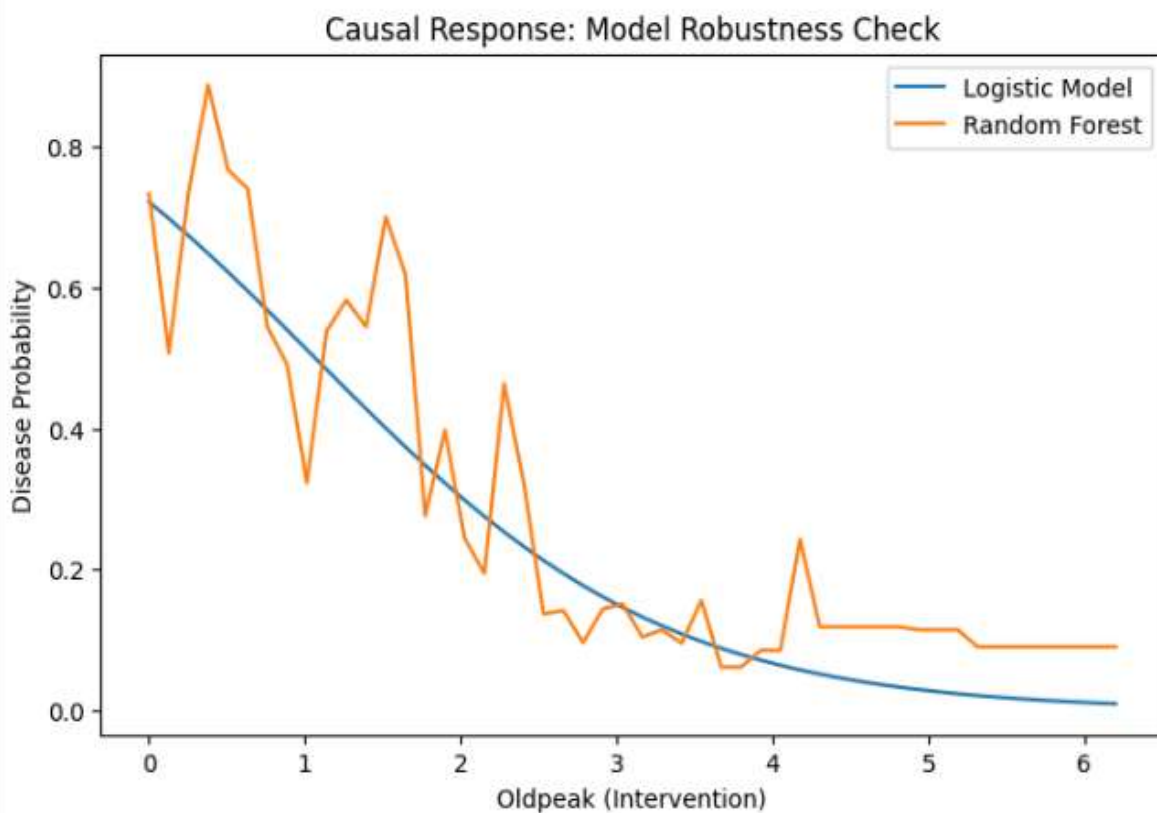
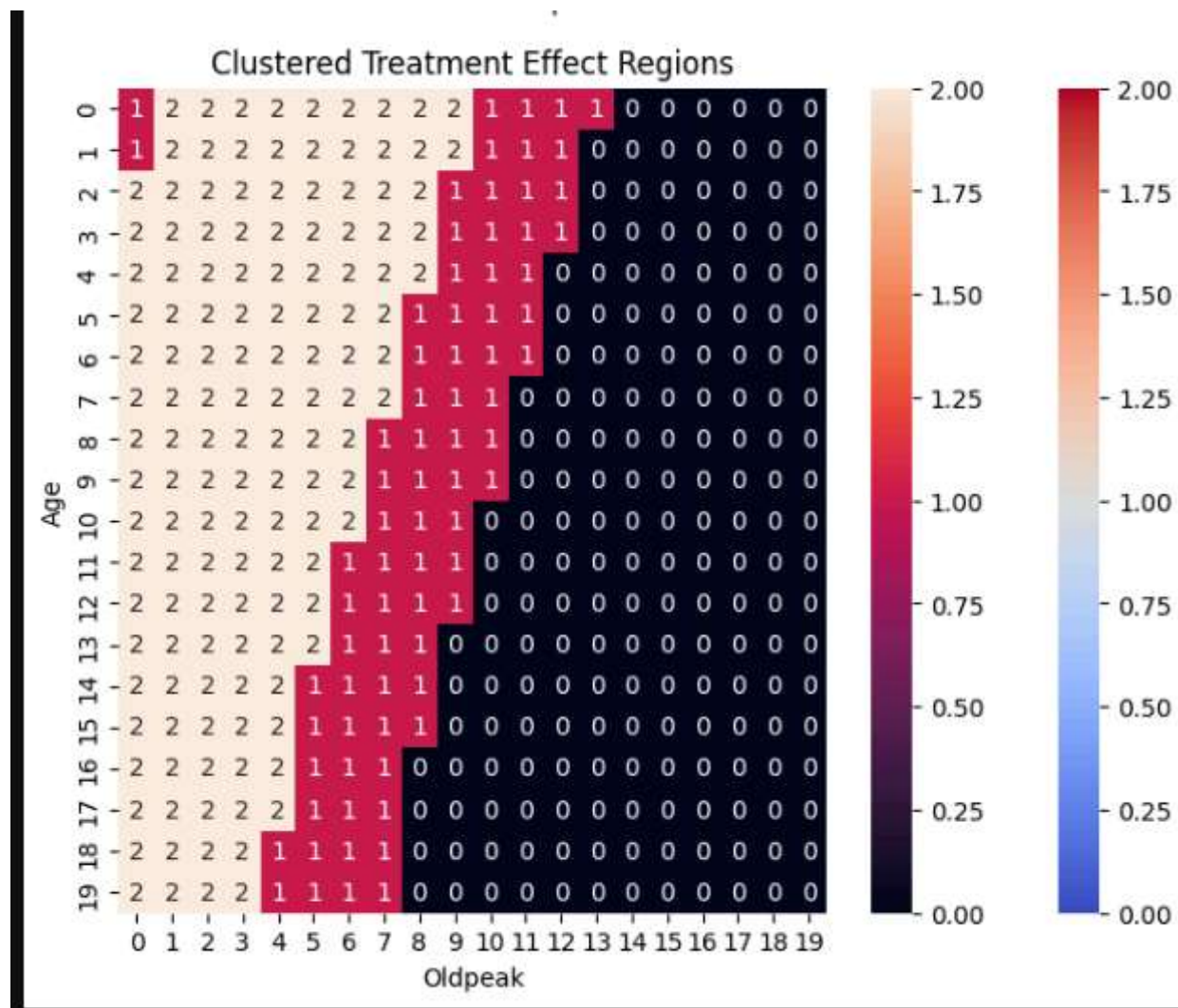


Figure 3: Patient-specific causal response showing diminishing effect of oldpeak.









Key Observations and Analytical Pitfalls :

During the analysis, several important statistical and causal insights were identified, highlighting limitations of naive modeling approaches:

- The relationship between oldpeak and heart disease is not consistent across all subgroups. While a negative correlation is

observed in most age groups, the Elder subgroup exhibits a positive correlation, indicating potential aggregation bias and Simpson's paradox.

- This apparent contradiction is largely driven by the small sample size and class imbalance in the Elder subgroup ($n \approx 20$ with high disease prevalence), demonstrating how correlation estimates can become unstable in low-variance populations.
- Linear regression was found to be unsuitable due to its inability to constrain predictions within probabilistic bounds $[0,1]$, making logistic regression more appropriate for this task.
- The assumption of homoscedasticity is violated in binary classification settings, as residuals are not evenly distributed but instead cluster at discrete values (0 and 1), further reducing the reliability of linear models.
- These observations reinforce the importance of using appropriate statistical models and highlight the need for causal reasoning over naive correlation-based interpretations.

KEY INSIGHTS :

Key Insights:

- Oldpeak exhibits a consistent negative causal relationship with heart disease probability.
- The treatment effect is strongest at lower oldpeak values, indicating that early-stage intervention yields the highest benefit.
- The effect diminishes as oldpeak increases, demonstrating a saturation behavior consistent with logistic response dynamics.

- Age significantly influences baseline disease risk but does not strongly modify the treatment effect of oldpeak, indicating weak interaction.
- Heterogeneous Treatment Effect analysis reveals that treatment effectiveness is localized, with clear regions of strong, medium, and weak impact.
- Causal analysis reveals insights that are not observable through standard correlation-based approaches, reinforcing the necessity of causal reasoning in real-world decision systems.

LIMITATIONS AND REAL-WORLD CONSIDERATIONS:

While this study provides valuable insights into causal relationships, several limitations must be acknowledged:

1. Observational Data:

The analysis is based on observational data rather than randomized controlled trials (RCTs). As a result, causal conclusions rely on assumptions and may be affected by unobserved confounders.

2. Limited Feature Set:

Only a subset of variables (primarily oldpeak and age) was used. Other important clinical factors may influence heart disease but were not included in the causal model.

3. Small Subgroup Sizes:

Certain subgroups, particularly the Elder age group, have limited sample sizes. This leads to unstable estimates and potential anomalies such as misleading subgroup correlations.

4. Model Assumptions:

The logistic regression model assumes a specific functional form (logistic relationship), which may not fully capture complex real-world biological interactions.

5. Synthetic Intervention Assumption:

The do-operator simulation assumes that intervening on oldpeak is feasible and independent, which may not reflect real-world medical interventions.

6. Medical Interpretation Limitation:

The findings of this analysis should not be interpreted as medically accurate conclusions. In real clinical settings, an increase in oldpeak is generally associated with higher heart disease risk. The inverse relationship observed in this dataset is likely due to dataset-specific patterns, confounding structures, or sampling bias.

7. External Validity:

The results may not generalize to broader populations, as the dataset may not represent diverse demographic or clinical conditions.

These limitations highlight the importance of combining data-driven causal analysis with domain expertise and real-world validation.

CONCLUSION :

This project demonstrates the critical distinction between correlation and causation in healthcare data analysis. While traditional machine learning models are effective at prediction, they often fail to capture the true causal relationships necessary for informed decision-making.

By incorporating causal inference techniques, including confounder adjustment, counterfactual reasoning, and intervention analysis, this study reveals that oldpeak exhibits a consistent causal effect on heart disease probability. Furthermore, the analysis highlights that this effect is not uniform across all conditions, showing strong impact at lower values and diminishing returns at higher levels, indicating a saturation behavior.

The use of individualized treatment effect estimation and heterogeneous treatment effect mapping provides deeper insights into how interventions may affect different patients, moving the analysis from population-level trends to personalized understanding.

Overall, this work emphasizes that transitioning from predictive modeling to causal reasoning is essential for building reliable, interpretable, and actionable systems, especially in high-stakes domains such as healthcare. It lays a foundation for more advanced causal modeling approaches and demonstrates the potential of data-driven decision support when grounded in causal principles.

This project demonstrates how causal thinking transforms data analysis from descriptive insight to actionable intelligence.

FUTURE WORK:

- Incorporating additional clinical variables to improve causal validity.
- Applying advanced causal methods such as propensity score matching or causal forests .
- Validating findings using real-world clinical datasets or experimental data .